

Siddharth Khare

Pre-Final year Undergraduate | Mathematics + Electronics | BITS Pilani

✉ siddharth.khare27@gmail.com | 📞 +91 9109656824 | [in](https://www.linkedin.com/in/siddharth-khare-27bp) [siddharth-khare-27bp](https://www.linkedin.com/in/siddharth-khare-27bp) | [github](https://www.github.com/siddharthk27) [siddharthk27](https://www.github.com/siddharthk27)

EDUCATION

Birla Institute of Technology and Science, Pilani

2027 (Dual Degree)

M.Sc. (Hons.), Mathematics and B.E., Electronics and Instrumentation

- **Relevant Coursework:** Computer Programming, Operating Systems, Machine Learning for Electronics Engineering, Statistical Inference and Applications, Applied Statistical Methods, Nonlinear Optimisation, Operations Research, Algebra, Graph Theory, Control Systems

PUBLICATIONS

1. **Siddharth Khare**, Jay Agarwal, Dhruv Kumar. “What do Lorentz Equivariant Jet taggers learn?” *ICML 2026 Workshop on AI for Physics (AI4Physics)* [\[link\]](#)
2. Sanyam Jain, **Siddharth Khare**, Jagat Sesh Challa. “KAHLT: Kolmogorov-Arnold Hybrid Latent Transformer - A Hybrid Transformer For Classification Tasks.” *Proceedings of the 18th International Conference on Agents and Artificial Intelligence (ICAART 2026)*. [\[link\]](#)
3. Pranav Deepak Tanna, **Siddharth Khare**, Sanyam Sanjay Jain, Shivodit Raj Vishnoi, Samay Roy, Jagat Sesh Challa. “Optimizing Liquid Neural Networks: A comparative study of LTCs and CFCs.” *Proceedings of the 2024 IEEE International Conference on Big Data (IEEE Big Data 2024)*. [\[link\]](#)
4. **Siddharth Khare**, Jay Agarwal, Dhruv Kumar. “Distilling the Knowledge of Lorentz Equivariant Transformers” (*in preparation*) [\[report\]](#)

WORK EXPERIENCE

Flipkart

May 2026 - July 2026

Data Science Intern, Search Team

Bangalore, India

- Built an end to end automated pipeline for identifying and diagnosing issues along every step of the Search Retrieval and Ranking system.
- Designed and built a profile-centric **LLM-as-a-judge** pipeline that replaces manual relevance annotation for search personalization, scoring product fit against users’ browse and purchase histories across **10,000+** products per evaluation run.
- Integrated Flipkart Search, Catalog and Price APIs for live retrieval and structured product enrichment; improved judgement quality using a multi-agent system for user preference extraction, query-aware attribute overrides, few-shot chain of thought and dynamic store-wise prompt rubrics.
- Automated the end-to-end labelling and evaluation workflow, reaching **70% agreement with human labels** for the llm while cutting down the labelling time from **45 mins per query for manual labelling to 18s per query**.

RESEARCH EXPERIENCE

PhantomGrad Lab

Nov 2025 – Present

Undergraduate Researcher (Guide: Dr. Dhruv Kumar)

BITS Pilani

- Developed an **interpretability framework for equivariant transformers** using linear probes, representation similarity, invariance tests and structured ablations to track task-relevant information across layers.
- Showed that symmetry-aware models suppress transformation-dependent nuisance information while preserving predictive features, and identified redundant representation pathways that could enable simpler architectures.
- Built a **knowledge-distillation pipeline** compressing a $\sim 1\text{M}$ -parameter physics-informed transformer into a **205K-parameter student**, matching its AUC (**0.9870**) with approximately **5 $\times$ fewer parameters**.
- Evaluated student architectures, loss functions, soft-label objectives, temperatures, intermediate-feature supervision and relational features; found that **architectural inductive bias** mattered most and transferred approximately **41% of the teacher’s robustness** to unseen transformations.

Postman Innovation Lab

Jan 2024 – Dec 2025

AI/ML Lead (Supervisor: Dr. Jagat Sesh Challa)

BITS Pilani

- Integrated Perceiver-Style Latent Attention with Kolmogorov Arnold Networks resulting in the **Kolmogorov Arnold Hybrid Transformer**, enabling non-linear feature processing with linear scalability.
- Evaluated these hybrid architectures on the **AG News** and **arXiv** datasets, achieving a peak accuracy of **91.43%**; work culminated in publication **KAHLT (ICAART 2026)**.
- Optimized **Liquid Time-Constant (LTC)** and **Closed-Form Continuous (CFC)** networks for behavior cloning tasks, achieving **93.74%** accuracy on Atari Breakout.
- Benchmarked **Liquid Neural Network (LNN)** performance against LSTM baselines across domains including **autonomous steering angle prediction** and GHI forecasting; work culminated in publication **LNN (IEEE Big Data 2024)**.

Indira Gandhi Centre for Atomic Research (IGCAR)

May 2024 – July 2024

Research Intern (Guide: Dr. Ajay Keshari)

Dept. of Atomic Energy, Gov. of India

- Developed **Data Acquisition (DAQ)** software in **LabVIEW** using **RS-232** protocols to interface with **calorimeter** instruments for high-precision thermal measurement.
- Designed an automated validation tool to compare experimental sensors against standard benchmarks, enabling robust **multi-day data logging** and measurement tracking.
- Facilitated the calculation of **isotopic concentrations** and sample age determination by integrating longitudinal heat-flux data into the automated measurement pipeline.

TEACHING EXPERIENCE

Teaching Assistant

Mar 2026 - June 2026

Operating Systems

BITS Pilani Digital

- Facilitated **real-time query resolution** for **1,300+ students** during live lectures, addressing core concepts in concurrency, memory management, and file systems.
- Evaluated academic assessments and programming assignments

PROJECTS

Shift Invariant Transformer for Side-channel analysis | [\[link\]](#)

Pytorch

- Implemented **Estranet** - a **shift invariant linear attention** transformer for side-channel analysis in pytorch
- Experimented with **Laplacian kernel** based attention against **Gaussian attention**, empirically verifying more stable training for Gaussian kernels.

SFuzz - Slice-based Fuzzing for RTOS (Replication Study) | [\[link\]](#)

Ghidra, Unicorn AFL, Angr

- Replicated a binary fuzzing system for Real-Time Operating Systems, successfully **reproducing vulnerability discovery results across 35 firmware samples**.
- Pipeline involved **static taint analysis** in Ghidra, **instruction-level CPU emulation** via Unicorn, and **concolic execution** with Angr.

Multi-Modal Product Price Estimation | [\[link\]](#)

CLIP, Regression

- Implemented a multi-modal solution for estimating product prices from images and text description by training a regression network on **fused Image + Text embeddings from CLIP**.
- Achieved a SMAPE score of 0.54 on the Amazon ML Challenge '25 dataset.

Recommendation Engine for ranking customer offers | [\[link\]](#)

PySpark, Pandas, DeepFM

- Engineered end-to-end **learning-to-rank** pipeline for predicting and ordering personalized reward offers using PySpark and Pandas on high-volume transactional and behavioral data from millions of Amex users.
- Built a **LambdaMart + DeepFM ensemble**, with hyperparameter tuning using Optuna, achieving a MAP@7 score of 0.416.

EXTRACURRICULARS

- Member of the Debating Society BITS Pilani, participated in nationwide inter-college Parliamentary Debates.
- Member of Astronomy Club, engaged in astronomy sessions and discussions on astrophysics.

TECHNICAL SKILLS

Languages/Tools: Python, C, NI Labview

Frameworks/Libraries: PyTorch, TensorFlow, LangChain